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UNIVERSITY OF AGRICULTURE, MULTAN

"Rabbi Zidni Ilma" — My Lord, increase me in knowledge

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DEPARTMENT OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

Faculty of Computing

PROGRAMMING FUNDAMENTALS · CS 303

Programming Fundamentals

CS-303

Assignment No. 1

Submitted By

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Program	BS Artificial Intelligence
Semester	1st
Section	A
Department	Data Science and Artificial Intelligence
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Lab no. 1

Task 1: Write a program that prints your name, roll number and program on separate lines.

Program:

main.cpp

Run

```cpp
#include <iostream>
#include <string>
using namespace std;
int main() {
    string name, rollno, program;
    cout << "Enter your name: ";
    getline(cin, name);
    cout << "Enter your Roll no: ";
    getline(cin, rollno);
    cout << "Enter your Program: ";
    getline(cin, program);
    cout << "Name: " << name << endl;
    cout << "Roll No: " << rollno << endl;
    cout << "Program: " << program << endl;
    return 0;
}
```

Output:

Output

```
Enter your name: Hassan Shahzad Qureshi
Enter your Roll no: 2477011
Enter your Program: BS AI (Artificial Intelligence)
Name: Hassan Shahzad Qureshi
Roll No: 2477011
Program: BS AI (Artificial Intelligence)
```

Task 2: Print a box of asterisks 5 rows high and 10 columns wide using cout statements.

Program:

main.cpp Run

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "Name: Hassan Shahzad\n";
    cout << "**********\n";
    cout << "**********\n";
    cout << "**********\n";
    cout << "**********\n";
    cout << "**********\n";
    return 0;
}
```

Output:

Output

```
Name: Hassan Shahzad
**********
**********
**********
**********
**********
```

Task 3: Record three different compiler errors and explain what caused each.

Program:

main.cpp Run

```cpp
#include <iostream>
using namespace std;
int main() {
    // Error 1: Missing semicolon
    // cout << "Hello"

    // Error 2: Undeclared identifier
    // count = 10;

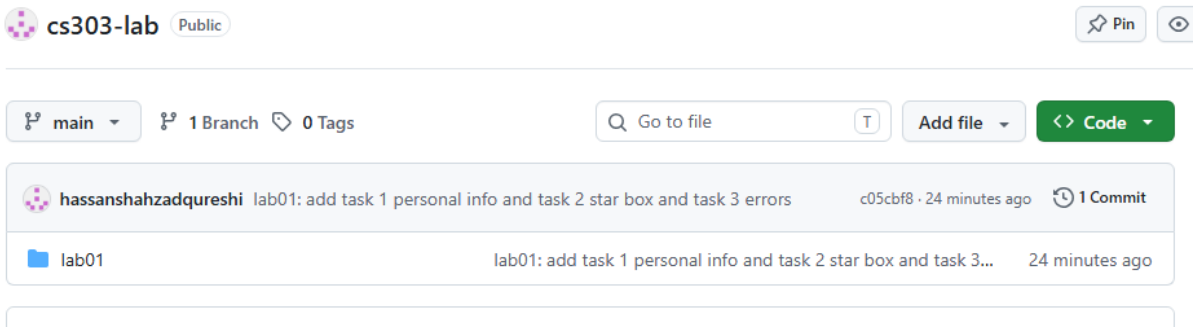
    // Error 3: Missing header or unresolved identifier (e.g., without #include <string>)
    // string s = "test"; // (fails if <string> is omitted in strict standard modes)
    // OR Type mismatch assignment:
    // int x = "Hello";
    cout << "Read comments above for the 3 documented errors.\n";
    return 0;
}
```

Output:

- **Error 1: Missing Semicolon (`expected ';' before '...'`)**
    - *Cause:* Semicolons terminate statements in C++. The parser expects a semicolon before encountering the next statement or closing brace.
- **Error 2: Undeclared Identifier (`'count' was not declared in this scope`)**
    - *Cause:* C++ is statically typed; every identifier must be declared with its type before use.
- **Error 3: Type Incompatible Assignment (`cannot convert 'const char*' to 'int' in initialization`)**
    - *Cause:* Attempting to assign a string literal to a primitive integer variable violates type safety rules.

Task 4: Push your repository to GitHub and share the link.

Link: https://github.com/hassanshahzadqureshi/cs303-lab/



Lab no. 2

Task 1: Read the length and width of a field in metres and print its area in square metres and in kanals (1 kanal = 505.857 m²).

Program:

main.cpp

Run

```cpp
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
    constexpr double kSquareMetresPerKanal = 505.857;
    double length{0.0};
    double width{0.0};
    cout << "Enter field length in metres: ";
    if (!(cin >> length) || length <= 0) {
        cerr << "Error: Invalid length input.\n";
        return 1;
    }
    cout << "Enter field width in metres: ";
    if (!(cin >> width) || width <= 0) {
        cerr << "Error: Invalid width input.\n";
        return 1;
    }
    double areaSqM = length * width;
    double areaKanals = areaSqM / kSquareMetresPerKanal;
    cout << fixed << setprecision(3);
    cout << "\n--- Field Area Results ---\n";
    cout << "Area in Square Metres: " << areaSqM << " m^2\n";
    cout << "Area in Kanals:        " << areaKanals << " kanals\n";
    return 0;
}
```

Output:

```
Output

Enter field length in metres: 123.5
Enter field width in metres: 124.5

--- Field Area Results ---
Area in Square Metres: 15375.750 m^2
Area in Kanals:        30.395 kanals
```

Task 2: Read temperature in Celsius and print it in Fahrenheit and Kelvin with two decimal places.

Program:

main.cpp — Run ▶

```cpp
// Read temperature in Celsius and print it in Fahrenheit and Kelvin with two decimal places.
#include <iomanip>
#include <iostream>
using namespace std;
int main() {
    double celsius{0.0};
    cout << "Enter temperature in Celsius: ";
    if (!(cin >> celsius)) {
        cerr << "Error: Invalid numeric input.\n";
        return 1;
    }
    double fahrenheit = (celsius * 9.0 / 5.0) + 32.0;
    double kelvin = celsius + 273.15;
    cout << fixed << setprecision(2);
    cout << "\n--- Temperature Conversion ---\n";
    cout << "Celsius:    " << celsius << " C\n";
    cout << "Fahrenheit: " << fahrenheit << " F\n";
    cout << "Kelvin:     " << kelvin << " K\n";
    return 0;
}
```

Output:

```
Output

Enter temperature in Celsius: 100

--- Temperature Conversion ---
Celsius:    100.00 C
Fahrenheit: 212.00 F
Kelvin:     373.15 K
```

Task 3: Read a number of seconds and print it as hours, minutes and seconds using / and %.

Program:

main.cpp — Run ▶

```cpp
// Read a number of seconds and print it as hours, minutes and seconds using / and %.
#include <iostream>
using namespace std;
int main() {
    long long totalSeconds{0};
    cout << "Enter total number of seconds: ";
    if (!(cin >> totalSeconds) || totalSeconds < 0) {
        cerr << "Error: Please enter a valid non-negative integer.\n";
        return 1;
    }
    long long hours = totalSeconds / 3600;
    long long remainingSeconds = totalSeconds % 3600;
    long long minutes = remainingSeconds / 60;
    long long seconds = remainingSeconds % 60;
    cout << totalSeconds << " seconds = "
    << hours << " hour(s), "
    << minutes << " minute(s), and "
    << seconds << " second(s).\n";
    return 0;
}
```

Output:

```
Output

Enter total number of seconds: 3665
3665 seconds = 1 hour(s), 1 minute(s), and 5 second(s).
```

Task 4: Demonstrate overflow by adding 1 to INT_MAX and explain the result.

Program:

main.cpp Run

```cpp
// Demonstrate overflow by adding 1 to INT_MAX and explain the result
#include <climits>
#include <iostream>
using namespace std;
int main() {
    int maxVal = INT_MAX;
    cout << "Value of INT_MAX: " << maxVal << "\n";

    // Adding 1 causes signed integer overflow
    int overflowedVal = maxVal + 1;
    cout << "Value of INT_MAX + 1: " << overflowedVal << "\n";
    int underflowedVal = overflowedVal - 1;
    cout << "Value of INT_MAX - 1: " << underflowedVal << "\n";
    return 0;
}
```

Output:

```
Output

Value of INT_MAX: 2147483647
Value of INT_MAX + 1: -2147483648
Value of INT_MAX - 1: 2147483647
```

Lab no. 3

Task 1: Read three numbers and print the largest using nested if.

Program:

task1.cpp

lab03 > task1.cpp

```cpp
// Read three numbers and print the largest using nested if.
#include <iostream>
using namespace std;
int main() {
    double a{0.0}, b{0.0}, c{0.0};
    cout << "Enter three numbers: ";
    if (!(cin >> a >> b >> c)) {
        cerr << "Error: Invalid input.\n";
        return 1;
    }
    double largest{0.0};
    // Using nested if structure
    if (a >= b) {
        if (a >= c) {
            largest = a;
        } else {
            largest = c;
        }
    } else {
        if (b >= c) {
            largest = b;
        } else {
            largest = c;
        }
    }
    cout << "The largest number is: " << largest << "\n";
    return 0;
}
```

Output:

Output

```
Enter three numbers: 10 20 30
The largest number is: 30
```

Task 2: Write a grade calculator for marks 0–100 using the grading scheme of the University semester rules.

Program:

```cpp
semester rules.
#include <iostream>
using namespace std;
int main() {
    double marks{0.0};
    cout << "Enter course marks (0 - 100): ";
    if (!(cin >> marks) || marks < 0.0 || marks > 100.0) {
        cerr << "Error: Invalid marks! Must be between 0 and 100.\n";
        return 1;
    }
    string grade;
    double gpa = 0.0;
    if (marks >= 85.0) {
        grade = "A";   gpa = 4.0;
    } else if (marks >= 80.0) {
        grade = "A-";  gpa = 3.7;
    } else if (marks >= 75.0) {
        grade = "B+";  gpa = 3.3;
    } else if (marks >= 70.0) {
        grade = "B";   gpa = 3.0;
    } else if (marks >= 65.0) {
        grade = "B-";  gpa = 2.7;
    } else if (marks >= 61.0) {
        grade = "C+";  gpa = 2.3;
    } else if (marks >= 58.0) {
        grade = "C";   gpa = 2.0;
    } else if (marks >= 50.0) {
        grade = "D";   gpa = 1.0;
    } else {
        grade = "F";   gpa = 0.0;
    }
    cout << "Marks: " << marks
         << " | Grade: " << grade
         << " | GPA Points: " << gpa << "\n";
    return 0;
}
```

Output:

```
Output

Enter course marks (0 - 100): 90
Marks: 90 | Grade: A | GPA Points: 4
```

Task 3: Write a menu-driven calculator (+, −, ×, ÷) using switch, handling division by zero.

Program:

```cpp
// lab03/task3_calculator.cpp
#include <iostream>
using namespace std;
int main() {
    double num1{0.0}, num2{0.0};
    char op{' '};
    cout << "---------------------------------\n";
    cout << "      Simple Arithmetic Calculator\n";
    cout << "---------------------------------\n";
    cout << "Enter first number: ";
    if (!(cin >> num1)) {
        cerr << "Error: Invalid first number.\n";
        return 1;
    }
    cout << "Enter an operator (+, -, *, /): ";
    cin >> op;
    cout << "Enter second number: ";
    if (!(cin >> num2)) {
        cerr << "Error: Invalid second number.\n";
        return 1;
    }
    switch (op) {
        case '+':
            cout << num1 << " + " << num2 << " = " << (num1 + num2) << "\n";
            break;
        case '-':
            cout << num1 << " - " << num2 << " = " << (num1 - num2) << "\n";
            break;
        case '*':
            cout << num1 << " * " << num2 << " = " << (num1 * num2) << "\n";
            break;
        case '/':
            if (num2 == 0.0) {
                cerr << "Error: Division by zero is undefined!\n";
            } else {
                cout << num1 << " / " << num2 << " = " << (num1 / num2) << "\n";
            }
            break;
        default:
            cerr << "Error: Unsupported operator '" << op << "'.\n";
            break;
    }
    return 0;
}
```

Output:

```
Output

=================================
      Simple Arithmetic Calculator
=================================
Enter first number: 10
Enter an operator (+, -, *, /): +
Enter second number: 10
10 + 10 = 20
```

Task 4: Determine whether a year is a leap year.

Program:

main.cpp

Run ▶

```cpp
// Determine whether a year is a leap year
#include <iostream>
using namespace std;
int main() {
    int year{0};
    cout << "Enter a year: ";
    if(!(cin >> year) || year < 0) {
        cerr << "Error: Invalid Year\n";
    }
    if (year % 4 == 0) {
        cout << year << " is a leap year.\n";
    } else {
        cout << year << " not a leap year.\n";
    }
    return 0;
}
```

Output:

Output

```
Enter a year: 2024
2024 is a leap year.
```

